

# Risk Management Field Manual

Memorize the whole deck the easy way: one story, the exact exam points, and a memory hook on every topic.

Module **MMC-01a**    Lecturer **Prof. Dr. M. Górka**    Source **3.pptx** · **73 slides**

**THD** · **NuW** · **AI for Smart Sensors**



## Meet Riku — your memory anchor

You and your friends are building Riku, a self-driving delivery robot. You're terrified it'll crash, hurt someone, or arrive late.

Every single slide just answers one question: “How do clever engineers stop Riku from going wrong — as early and cheaply as possible?” Keep Riku in your head and the chapter tells itself.

★ **MASTER KEY** → “**Detectives Always Eat Cold Mexican Donuts**” =  
**Detect** · **Analyse** · **Evaluate** · **Control** · **Monitor** · **Document**

# See the skeleton before the bones

Risk Management is step 3 of System Engineering:

1 Basics → 2 Requirements → **3 RISK ★** → 4 Change → 5 Ideation → (6 Design) →  
(7 Test) → (8 Project Setup)

The chapter solves risk in a fixed order. Learn this one silly sentence and you already own the structure of the exam — everything else just bolts onto one of these six words:



**“Detectives Always Eat Cold Mexican Donuts.”**

**Detect → Analyse → Evaluate → Control → Monitor → Document.**

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## The Memory Movie

Don't memorize sentences — your brain hates random letters. Instead **close your eyes and play this silly 6-scene movie** of Riku's bad day. Each picture is the point, so when you replay the movie, the exam answer just appears. Read it 3 times, then try to replay it with your eyes shut.



### SCENE 1

A detective spots a **banana peel** on Riku's path. → **DETECT** the risk.



### SCENE 2

He drops the peel under a **microscope** to study it. → **ANALYSE** it.



### SCENE 3

He plops it on a **kitchen scale**: “how slippery? × how often is it here?” → **EVALUATE** = **Severity** × **Occurrence**.



### SCENE 4

He grabs Riku's **game controller** — a menu of 4 **buttons** pops up: 🧱 🌂 🧑 🧨. → **CONTROL** it (4 options below).



### SCENE 5

He leans back and watches Riku on **CCTV screens** to see if it slips. → **MONITOR** it.



### SCENE 6

He scribbles the whole story into a **logbook**. → **DOCUMENT** it.



Replay in one breath: **spots** → **studies** → **weighs** → **grabs the controller** → **watches the screens** → **writes the logbook**. That's the whole chapter's skeleton — no letters needed.



**The 4 buttons on the controller (how to CONTROL a risk)**

Picture standing in your doorway in the rain, choosing one button:



### Wall = Avoid

Build a wall so the rain never reaches you.  
**Reduce occurrence to 0** (e.g. Poka Yoke).



### Umbrella = Prevent

You still go out, but stay drier. **Reduce occurrence or severity** (e.g. PDCA).



### Brother = Transfer

Push someone who copes better into the rain.  
**ECU → chassis.**



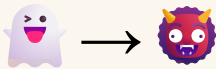
### Shrug = Accept

Just get a little wet. **Only if it's a drizzle** (low × low).



## Picture-hooks for the tricky lists

Same trick for every list that wouldn't stick — turn it into one strong, weird image:



### Risk vs Problem

A floating ghost “?” = a risk (not yet, <100%). It drops to the floor and becomes a **monster** = a problem (already happened).



### PDCA

Bake cookies on a **spinning plate**: **Plan** recipe → **Do** bake → **Check** taste → **Act** fix next batch. Round forever.



### 6 ways to DETECT

Walk into a room: a  **dashboard** (KPI), an old  **diary** (LeLe/lessons), a kid doing a  **headstand**, a  **fortune-teller** (Delphi), a  **workshop bench**, a  **hazard sign** (PAAG/HAZOP).



### RPV = “how SORe”

Riku trips and is **SORE**: how bad the bruise (Severity) × how Often he trips (Occurrence) × can he see it coming (Recognizability). Each /10.



### FMEA family

A plain robot (FMEA) puts on a 👑 **crown** to rank what's critical (FMECA) → then a 🩺 stethoscope to self-diagnose (FMEDA).



### Rule of Ten

A snowball at the hill-top: 1€ → 10€ → 100€ → 1000€, smashing the customer's house. **Catch it at the TOP (planning).**



### Where risks hide (9→3)

**Product box** (Technical, Compliance) · **Money/Time box** (Finance, Time, Market, Currency) · **People/Planet box** (Social, PM, Environmental).



### SC car badges

A BMW wears 3 badges L·S·F (Legal, Safety, Function); a VW wears 2 badges D·S (Documented, Safety). Daimler = DS/DZ.



### Standards anchor

A race-car with number plate “26262” = ISO 26262 (automotive FMEA). Anchor on the car, the rest cluster around it.



### Delphi vs Headstand

**Delphi** = silent hooded oracles, answer alone, repeat, average. **Headstand** = flip the question (“how to make Riku fail FAST?”), then flip the answers back.



**How to actually lock it in:** (1) read a scene, (2) **shut your eyes and re-see the picture**, (3) say the real meaning out loud. Do the 6-scene movie + 10 picture-hooks twice today and once tomorrow — then test yourself with the **flashcards**. Seeing beats reading; testing beats re-reading.

# Taught like a story — with the points you must write

## 01 What *is* a risk?

### STORY

Before Riku even moves you ask, “what could go wrong?” The battery *might* die; it *might* rain. Hasn't happened yet → those are **risks**. The instant Riku actually crashes, it stops being a risk and becomes a **problem** — too late to prevent.

### WRITE IN EXAM — DEFINITION OF RISK

- “**Risk is the effect of uncertainty**” — definition from **ISO 9001**.
- RM's job: **sort & structure risks** in a structured way (→ PDCA).
- Risks = events/circumstances that can **prevent reaching the project goal** (probability < 100%).
- A risk that **already occurred** → a “**problem**” (treated differently).
- Separate **general company risks** from **project-specific risks**.

 Risk = “**not yet**” · Problem = “**already**”. Probability under 100% = still a risk.

## 02 PDCA — the engineer's heartbeat

### STORY

How do you do anything well? Like baking cookies: **Plan** the recipe → **Do** the baking → **Check** the taste → **Act** to improve the next batch. Engineers spin this loop forever. From ISO 9001; also the **Deming / Shewhart** cycle.

#### WRITE IN EXAM — PDCA

- **Plan** — plan process/product; set goals & resources; address risks & opportunities.
- **Do** — implement the plan.
- **Check** — monitor & measure the process / target condition.
- **Act** — take action to improve performance.
- Method used in “Plan”: **FMEA**.



P-D-C-A = “Please Don't Crash Again.” It's a **circle**, never a straight line.

### 03 Three clever truths (general principles)

#### STORY

Three surprising facts examiners love: (1) risk has a good twin — an **opportunity**; (2) people fear reporting bad news, so build an **anonymous “speak-up line”**; (3) money adds up but **risks don't** — you must combine them with maths like **Monte Carlo**.

#### WRITE IN EXAM

- Opportunities can be recorded as **positive events**.
- Set up an **anonymous “speak-up line”** for undiscovered risks/violations.
- **Risks do not add up** → aggregate via **models & simulations (Monte Carlo)**.



O-S-M → Opportunity · Speak-up · Monte-Carlo (risks don't add).

### 04 Don't put all eggs in one basket (diversification)

#### STORY

Bet all your pocket money on one lottery ticket = super risky. Spread it across different, *unrelated* things = safer for the same reward. Companies do the same with projects. To make Riku's 50%-reliable crash sensor safer, you must add something **different**, not a copy.

#### WRITE IN EXAM

- Diversify — “not all in one basket”: **asset** (different products) & **time** diversification (idea by **Markowitz**, Nobel Prize).
- Optimum portfolios lie **on the efficiency curve**; below it = **inefficient**.
- To reduce risk: **diversified & non-correlating** solutions.
- ECU example — **✗ wrong**: add a 2nd **identical** ECU (increases failure). **✓ right**: add a **mechanical** crash detector.



Two of the **same** = double trouble. Two **different** = real safety.

## 05 The rulebook (standards)

#### STORY

You can't just say you manage risk — big standards *force* you. **ISO 9001** demands “risk-based thinking”; **ISO 31000** is the dedicated risk standard; companies keep a money-jar (**provisions**) to pay if risks hit; and the **boss** is responsible.

#### WRITE IN EXAM

- **ISO 9001** → “**risk-based thinking**” (use opportunities, prevent unwanted outcomes).
- **ISO 31000** → “Risk Management”.
- Form **provisions (Rückstellungen)** for company & product risks → linked to **quality management**.
- Company-level RM = a **managerial task**.



“9001 thinks, 31000 manages, the jar pays, the boss owns it.”

## 06 Risks come from everywhere (levels affected)

#### STORY

Riku can fail in many directions. Turn 9 scary items into **3 buckets**: Product · Money/Time · People/Planet.



#### WRITE IN EXAM – 9 LEVELS

- **Product:** Technical/product · Compliance
- **Money/Time:** Finances/costs · Time · Market · Currency
- **People/Planet:** Social · Project-management · Environmental
- Also: **6 levels of risk management** by **Gleißner & Mott**; German law **KonTraG** forces management to do risk analysis.



**3 buckets** → **Product · Money/Time · People/Planet**. Remember the buckets, the items fall out.

## 07 Targets of Risk Management (master list)

#### STORY

This is the D-A-E-C-M-D skeleton, now with **Control** opened up. Controlling a risk = exactly 4 choices, like dealing with rain: **Avoid** (stay in), **Prevent** (umbrella), **Transfer** (send your brother), **Accept** (get a bit wet).

#### WRITE IN EXAM

- **Targets:** Detect → Analyse → Evaluate → **Control** → Monitor → Document & communicate.
- **Control options:** Avoid · Prevent · Transfer · Accept.



Skeleton = “**Detectives Always Eat Cold Mexican Donuts.**” Control = **A-P-T-A** (rain). Matches the classic **4 T's**: Terminate, Treat, Transfer, Tolerate.

## 08 DETECT — finding risks

#### STORY

First, hunt for risks **regularly & systematically**, looking *inside* and *outside* the company. A slow stable company checks rarely; a fast-changing one checks often.

#### WRITE IN EXAM

- Regular & systematic **internal and external** analysis of developments/events/changes; **frequency depends on volatility**.
- **6 methods to detect:** KPI changes · LeLe (lessons learned) · Headstand/Flip-Flop · Delphi · Risk workshop · PAAG/HAZOP.
- **Safety detection methods:** ALARP · FMEA · DRBFM.

 “Kids Love Hunting Down Risky Problems” → KPI · LeLe · Headstand · Delphi · Risk-workshop · PAAG.

### 08a Head-stand / Flip-Flop method

#### STORY

Stand the question on its head! Instead of “how do we make Riku succeed?” ask “**how could we make Riku fail as FAST as possible?**” — kids find this funnier, so more ideas pop out. Then flip the silly answers back into real safety actions.

#### 4 STEPS

- (1) **Invert the thesis** → (2) **find solutions** for the flipped thesis → (3) **invert those solutions** → (4) **define actions**.

### 08b Delphi method

#### STORY

Ask many experts — but keep them in **separate rooms** so nobody copies or bullies. Everyone answers **anonymously**; you reveal the big disagreements, ask again, repeat until they roughly agree, then take the **average**. (Named after the ancient Oracle of Delphi.)

#### 6 STEPS

- (1) leader explains topic & targets → (2) worksheet to each expert, **no discussion** → (3) **anonymous** hand-in + evaluate → (4) share **major deviations** + comments, redistribute → (5) **repeat 2–4** until consolidated → (6) take the **average**.

## 09 EVALUATE — scoring the risk

### STORY

Every risk gets **two scores**, like rating a scary movie: **Severity** (how bad) and **Occurrence** (how likely). They say “occurrence,” NOT “probability,” because it's a *human guess*, not a calculation. Combine the two → a **risk matrix** telling you what to fear first.

### WRITE IN EXAM

- Two properties per risk: **Severity** + **Occurrence/likelihood** (subjective; *not* “probability”).
- Combine → **risk matrix** to **prioritise**; keep an updated **risk list** (PM / risk manager).
- Judge **impact** via: **historical data** · **probability distribution** · **Monte Carlo simulation**.

 **Severity × Occurrence = Risk Matrix.** “How bad × how likely.” Low number = mild · high = nasty · 0 = gone · top = already happened.

## 10 CONTROL — the 5 ways to handle it

### STORY

Rain again, plus a bonus tool: **Avoid** (ban it, occurrence→0; use barriers like Poka Yoke), **Poka Yoke** (make wrong physically impossible — like a SIM that fits one way), **Prevent** (umbrella), **Transfer** (give it to who copes better), **Accept** (shrug — only if small).

### WRITE IN EXAM

- **Avoidance:** reduce occurrence to **0** — ban activities, remove features, **technical barriers** (Poka Yoke).
- **Poka Yoke:** Japanese “**avoid silly mistakes**”; immediate error detection; **key-lock principle**.
- **Prevention:** reduce occurrence OR mitigate severity, using QM methods (e.g. **PDCA**).
- **Transfer:** shift the risk to who handles it best — e.g. crash **ECU** → **chassis**.
- **Acceptance:** consciously accept — **only if occurrence AND impact are low**.



Avoid · Poka-Yoke · Prevent · Transfer · Accept = ban · mistake-proof · umbrella · chassis · shrug.

## 11 FMEA — the star of the exam ★

### 🧑 STORY

FMEA is a giant “what could break, and what happens then?” checklist that a team of experts fills in **early**, before Riku is even built — so failures get caught in advance.

### 📄 WRITE IN EXAM — FMEA BASICS

- **FMEA = Failure Mode and Effects Analysis** (Ger. *Fehlermöglichkeiten- und Einflussanalyse*) — a **reliability-improvement method**.
- Rates defects by **Severity, Probability, Recognizability**.
- **Goals:** spot **systematic & random errors before they occur** · derive **detection & prevention** · improve **successor products**.
- **Types:** System FMEA · **D-FMEA** (Design, customer focus) · **P-FMEA** (Process). D & P must be **linked** (critical design feature = an **SC** checked in the process).
- **Standards:** Automotive **ISO 26262** · Industry IEC 61508 / ISO 13849 · Aerospace DO-254 / ARP 4761.
- **Process:** prepared **early**, in **expert workshops**, built **from cause → error**, then **verify the assumed probabilities**.
- **Living document:** maintain until **EOP** (better **EOS**); **update on every product change** (classic field-failure cause!); old FMEA feeds next project (**LeLe**).



Family: A → +C(riticality)=FMECA → +D(iagnostics)=FMEDA. “Add a letter to upgrade.” · Standards anchor: “cars = 26262.”

### 11a Rule of Ten 10

### 🧑 STORY

The longer a mistake hides, the more it costs — **×10 at every stage**. Catch Riku's bug while planning = 1€; let the customer find it = 1000€+. So fix things **EARLY**.

#### WRITE IN EXAM

- Cost  $\times 10$  per stage: **Planning 1€**  $\rightarrow$  **Development 10€**  $\rightarrow$  **Assembly 100€**  $\rightarrow$  **Final acceptance 1000€**  $\rightarrow$  **Customer**. Focus on **planning & development**.



Every stage you're late, **add a zero**:  $1 \rightarrow 10 \rightarrow 100 \rightarrow 1000$ .

## 11b FMECA & FMEDA

#### WRITE IN EXAM

- **FMECA** = FMEA + **Criticality** metric  $\rightarrow$  enables **prioritisation** (usually what “FMEA” means in practice).
- **FMEDA** = FMEA + **Diagnostics**; **multiple-error analysis**; automotive electronics; counts how **self-detection/correction** lowers the failure rate; adds **quantifiable error data**; good for **complex systems**.

## 12 RPV — the FMEA score ( $S \times O \times R$ )

#### STORY

To rank each failure, FMEA gives **three marks out of 10** and multiplies them: how **bad** (Severity), how **often** (Occurrence), can our design **catch** it (Recognizability). Big number = fix first.

#### WRITE IN EXAM

- **RPV** = **Severity**  $\times$  **Probability(Occurrence)**  $\times$  **Recognizability**, each rated **1–10**.
- Watch the flip: Recognizability **1 = best** ( $\geq 99\%$  detected), **10 = worst** ( $< 10\%$  detected).



“**S  $\times$  O  $\times$  R**” = “**how SORe will it be?**” = how Bad  $\times$  how Often  $\times$  can we Recognize.  
(Classic FMEA **S  $\times$  O  $\times$  D** = **RPN**; Recognizability = Detection.)

## 12a Special Characteristic (SC)

#### STORY

If a feature is *super severe*, stamp it SC on the drawing — a red “MUST CHECK” sticker. Every SC is measured in production. But don't sticker *everything* (too costly).

#### WRITE IN EXAM

- High severity → **Special Characteristic (SC)**, marked on drawings, **each SC checked/measured**; can be a **legal/regulatory** requirement; don't over-use (cost!).
- **Automotive variants**: **BMW** = L/S/F (Legal/Safety/Function) · **Daimler** = DS/DZ · **VW** = D/S (Documented/Safety).



SC = red “MUST-MEASURE” sticker. BMW has 3 letters (LSF), VW has 2 (DS).

### 13 Loose ends + Summary

#### WRITE IN EXAM

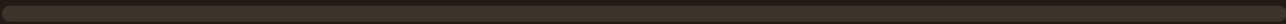
- **Software** to track risks: **ClickUp** · **cplace** · **Jira** → key point: **consistent linking** with up/downstream processes.
- Worked **example**: bicycle **cable-pull** FMEA (*341\_Example\_FMEA\_Bicycle*).
- **Summary**: RM **avoids greater damage** · identify risks **as early as possible** (Rule of Ten) · rate by **severity & occurrence** in a **risk list** · analyse products with **FMEA** & derive **SCs**.



**Walter Scheel**: “Nothing happens without risk, but without taking risks also nothing happens.” → manage risk, but still **take risks to move forward**.

YOUR PROGRESS

0 / 13 topics learned



# Flashcards

Read the question, say the answer *out loud*, then flip. Cover the deck and run it twice — that's how it sticks.

## All flashcards (Q/A)

**1. What is the definition of "risk" (ISO 9001)?**

"Risk is the effect of uncertainty." An event that can prevent reaching the project goal (probability < 100%).

**2. Risk vs problem?**

Risk = NOT happened yet (<100%). Problem = ALREADY occurred — handled differently.

**3. Two kinds of risk to keep separate?**

General company risks vs project-specific risks.

**4. PDCA + other names?**

Plan-Do-Check-Act. Also the Deming / Shewhart cycle (ISO 9001).

**5. PDCA — Plan?**

Plan process/product, set goals & resources, address risks & opportunities.

**6. PDCA — Do / Check / Act?**

Do = implement; Check = monitor & measure; Act = improve performance.

**7. Method used in the PDCA Plan phase?**

FMEA.

**8. Can a risk be positive?**

Yes — opportunities recorded as positive events.

**9. What is a speak-up line?**

An anonymous channel to report undiscovered risks & violations.

**10. Do risks add up like costs?**

No — aggregate them with models/simulations (Monte Carlo).

**11. Reduce risk through diversification?**

Diversified & NON-correlating solutions; asset & time diversification (Markowitz).

**12. 50% ECU — wrong vs right fix?**

Wrong: 2nd identical ECU (raises failure). Right: a different, mechanical detector.

**13. Efficiency curve meaning?**

Optimum portfolios on the curve; none above; below = inefficient.

**14. ISO 9001 risk concept?**

"Risk-based thinking" — use opportunities, prevent unwanted outcomes.

**15. ISO 31000?**

The dedicated Risk Management standard.

**16. Provisions (Rückstellungen)?**

Money set aside for company/product risks; linked to quality management.

**17. Company-level RM is what task?**

A managerial task.

**18. 9 levels risks affect?**

Technical, Compliance, Finances, Time, Market, Currency, Social, Project-management, Environmental.

**19. Who defined the 6 levels of RM?**

Gleißner & Mott.

**20. KonTraG?**

German law obliging management to perform risk analysis.

**21. The 6 targets/steps of RM?**

Detect → Analyse → Evaluate → Control → Monitor → Document & communicate.

**22. 4 options when CONTROLLING a risk?**

Avoid · Prevent · Transfer · Accept.

**23. What does Detect involve?**

Regular & systematic internal AND external analysis; frequency depends on volatility.



**24. 6 methods to detect risks?**

KPI changes, LeLe, Headstand/Flip-Flop, Delphi, Risk workshop, PAAG/HAZOP.

**25. 3 safety detection methods?**

ALARP, FMEA, DRBFM.

**26. Head-stand method — 4 steps?**

Invert thesis → find solutions → invert solutions → define actions.

**27. Delphi method key idea?**

Experts answer anonymously, no discussion, iterate on deviations, take the average.

**28. Evaluate: which two properties?**

Severity (how bad) + Occurrence/likelihood (how likely; subjective, not 'probability').

**29. Severity × Occurrence gives?**

A risk matrix to prioritise; risks kept in a risk list (PM/risk manager).

**30. 3 ways to assess impact?**

Historical data, probability distribution, Monte Carlo simulation.

**31. Severity & Occurrence scale ranges?**

Severity 1(low)→4(very high). Occurrence 0(0%)→5(100%=occurred).

**32. Risk Avoidance?**

Reduce occurrence to 0 — ban activities, remove features, Poka Yoke barriers.

**33. Poka Yoke?**

Japanese 'avoid silly mistakes' — mistake-proofing; key-lock principle.

**34. Risk Prevention?**

Reduce occurrence OR severity using QM methods (PDCA).

**35. Risk Transfer?**

Shift the risk to who handles it best — e.g. crash ECU → chassis.

**36. Risk Acceptance?**

Consciously accept — only if occurrence AND impact are low.

**37. FMEA — full form & what?**

Failure Mode and Effects Analysis — a reliability-improvement method.

**38. FMEA rates defects by?**

Severity, Probability (Occurrence), Recognizability.

**39. FMEA goals?**

Spot systematic & random errors before they occur; derive detection & prevention; improve successor products.

**40. FMECA?**

FMEA + Criticality metric → enables prioritisation.

**41. FMEDA?**

FMEA + Diagnostics; multiple-error analysis; automotive electronics.

**42. 3 types of FMEA?**

System, D-FMEA (Design), P-FMEA (Process).

**43. D-FMEA vs P-FMEA?**

D = design/customer focus (avoid systemic errors); P = production-process weaknesses; must be linked.

**44. FMEA standards by domain?**

Automotive ISO 26262; Industry IEC 61508 / ISO 13849; Aerospace DO-254 / ARP 4761.

**45. FMEA process?**

Early, in expert workshops, from cause→error, then verify assumed probabilities.

**46. Rule of Ten?**

Cost ×10 per stage: Planning 1€→Dev 10€→Assembly 100€→Final 1000€→Customer.

**47. FMEA maintained until?**

EOP (better EOS); update on product changes; old FMEA feeds next project (LeLe).

**48. How is RPV calculated?**

$RPV = \text{Severity} \times \text{Probability}(\text{Occurrence}) \times \text{Recognizability}$ , each 1–10.

**49. Recognizability scale direction?**

1 = surely recognized ( $\geq 99\%$ ); 10 = certainly not recognized ( $< 10\%$ ).

**50. Special Characteristic (SC)?**

High-severity feature marked on drawings; each SC measured in production; can be legal.

**51. SC automotive variants?**

BMW L/S/F; Daimler DS/DZ; VW D/S.

**52. System FMEA?**

Whole system + subsystems; focuses on interfaces; finds random & systemic errors.

**53. Software to track risks?**

ClickUp, cplace, Jira — with consistent linking up/downstream.

**54. Summary core message?**

Avoid damage; identify early (Rule of Ten); rate by severity & occurrence; use FMEA + SC.

**55. Walter Scheel quote?**

"Nothing happens without risk, but without taking risks also nothing happens."

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## 3 Practice Questions + model answers

Try to answer on paper before clicking **Reveal**. The model answers show the exact points an examiner rewards.

**Q1 Describe the risk-management process. Name and explain its main steps, and the options available when *controlling* a risk.**

TYPE: STRUCTURED / THEORY · ~10 MARKS

REVEAL MODEL ANSWER

### ✓ MODEL ANSWER

Risk management follows a fixed cycle: **Detect** → **Analyse** → **Evaluate** → **Control** → **Monitor** → **Document & communicate** (underpinned by the PDCA / Deming cycle from ISO 9001).

- **Detect:** regular & systematic internal and external analysis (KPI changes, lessons learned, head-stand method, Delphi, risk workshops, PAAG/HAZOP).
- **Analyse & Evaluate:** score each risk by **severity** and **occurrence/likelihood** and place it in a **risk matrix** / risk list to prioritise.
- **Control** — four options: **Avoid** (reduce occurrence to 0, e.g. Poka Yoke), **Prevent** (reduce occurrence or severity, e.g. via PDCA), **Transfer** (shift to who handles it best, e.g. ECU→chassis), **Accept** (consciously, only if occurrence and impact are low).
- **Monitor:** check whether risks materialise; **Document & communicate** and define the RM organisation.

Distinguish a **risk** (not yet occurred, <100%) from a **problem** (already occurred). Note risks **do not add up** — aggregate via models (Monte Carlo).

**Q2**

**Explain FMEA. Distinguish FMEA, FMECA and FMEDA, state how the Risk Priority Value (RPV) is calculated, and explain the “Rule of Ten”.**

TYPE: THEORY / DEFINITIONS · ~12 MARKS

REVEAL MODEL ANSWER

✓ MODEL ANSWER

**FMEA (Failure Mode and Effects Analysis)** is a reliability-improvement method that, early and in cross-domain expert workshops, lists how a product can fail and what the effects are. It works **from cause to error** and aims to recognise systematic & random errors **before they occur**, derive detection & prevention measures, and improve successor products. It is a **living document** maintained until EOP/EOS.

- **FMEA → FMECA:** adds a **Criticality** metric → enables prioritisation.
- **FMEA → FMEDA:** adds **Diagnostics** (multiple-error analysis, automotive electronics) — credits self-detection that lowers the failure rate.

**RPV = Severity × Probability(Occurrence) × Recognizability**, each rated 1–10 (for Recognizability, 1 = surely detected, 10 = not detected). A high RPV means act first.

**Rule of Ten:** the later an error is found, the higher the cost — roughly **×10 per stage** (Planning 1€ → Development 10€ → Assembly 100€ → Final acceptance 1000€ → Customer). Therefore effort must focus on **planning & development**.

**Q3**

**Riku, an autonomous delivery robot, has an obstacle-detection sensor that is only 50% reliable. (a) Use diversification to improve it — and say what would be WRONG. (b) Classify two risks with severity & occurrence. (c) Name two ways to detect risks.**

TYPE: APPLICATION · ~8 MARKS

REVEAL MODEL ANSWER

✓ MODEL ANSWER

(a) Reliability is improved by **diversified, non-correlating** solutions. **Wrong:** add a second **identical** sensor — same solution to the same problem; it even raises the failure rate. **Right:** add a **different** detection principle (e.g. a mechanical/ultrasonic detector) so the two don't fail together. (Markowitz — don't put all eggs in one basket.)

(b) Examples (Severity 1–4, Occurrence 0–5 → risk matrix):

- Sensor briefly blinded by sunlight → frequent but mild: **Occurrence high, Severity low** → **medium**.
- Robot fails to brake and hits a pedestrian → rare but catastrophic: **Occurrence low, Severity very high** → **high**.

(c) Any two of: **Head-stand/Flip-Flop method**, **Delphi method**, KPI-change monitoring, lessons learned (LeLe), risk workshop, PAAG/HAZOP, or safety methods ALARP / FMEA / DRBFM. (*Head-stand: ask “how could we make Riku crash as fast as possible?”, then invert the answers into safety actions.*)

# One-screen cheat-card

|                 |                                                  |                                             |
|-----------------|--------------------------------------------------|---------------------------------------------|
| Skeleton        | Detect·Analyse·Evaluate·Control·Monitor·Document | “Detectives Always Eat Cold Mexican Donuts” |
| Control 4       | Avoid·Prevent·Transfer·Accept                    | Rain: A-P-T-A (= 4 T's)                     |
| Risk vs problem | not-yet vs already                               | <100% = risk                                |
| PDCA            | Plan·Do·Check·Act                                | “Please Don't Crash Again” (circle)         |
| Detect methods  | KPI·LeLe·Headstand·Delphi·Risk-workshop·PAAG     | “Kids Love Hunting Down Risky Problems”     |
| Evaluate        | Severity × Occurrence → matrix                   | bad × likely                                |
| FMEA family     | A → +C(rit) → +D(iag)                            | add a letter to upgrade                     |
| RPV             | Severity × Occurrence × Recognizability          | “how SORe” (= S×O×D)                        |
| Rule of Ten     | ×10/stage, 1→1000€                               | each stage late = +1 zero                   |
| Diversify       | different & non-correlating                      | 2 same = double trouble                     |
| SC              | high severity → must-measure                     | BMW=LSF, VW=DS                              |
| Standards       | cars = ISO 26262                                 | 9001 thinks · 31000 manages                 |
| Head-stand      | flip → solve → flip back → act                   | stand it on its head                        |
| Delphi          | silent · anonymous · repeat · average            | Oracle of Delphi                            |
| Levels affected | Product / Money-Time / People-Planet             | 3 buckets                                   |

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